



INTERNATIONAL GRAMMAR SCHOOL
Concordia per Diversitatem

Name:.....

INTERNATIONAL GRAMMAR SCHOOL

MATHEMATICS

Extension 2

YEAR 12

TRIAL EXAMINATION

31st JULY, 2001

**Time allowed ---3 hours
(Plus 5 minutes reading time)**

DIRECTIONS TO CANDIDATES

- Attempt **ALL** eight questions.
- ALL questions are of equal value. (8 @ 15 marks = 120 marks)
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a *new page*. Number each question clearly.
- Label each page with your name.
- A table of Standard Integrals is attached.

QUESTION 1 (Start a new page)

MARKS

a) Find $\int \frac{\sec^2 x}{\sqrt{1 - \tan^2 x}} dx$

2

b) i) Find a, b and c such that

2

$$\frac{16}{(x^2 + 4)(2 - x)} = \frac{ax + b}{x^2 + 4} + \frac{c}{2 - x}$$

ii) Find $\int \frac{16}{(x^2 + 4)(2 - x)} dx$

2

c) Find $\int \frac{\ln x}{x^2} dx$

4

d) Use the substitution $t = \tan \frac{\theta}{2}$ to show that

5

$$\int_0^{\pi} \frac{d\theta}{4 \sin \theta - 2 \cos \theta + 6} = \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right)$$

QUESTION 2 (Start a new page)

MARKS

a) The complex number Z moves such that $\operatorname{Im} \left(\frac{1}{\bar{Z} - i} \right) = 1$.

3

Show that the locus of Z is a circle and find its centre and radius.

b) i) Find the square root of the complex number $5 - 12i$

2

ii) Given that $Z = \frac{1 + \sqrt{5 - 12i}}{2 + 2i}$ and is purely imaginary, find Z^{400}

2

c) i) Shade the region in the argand diagram containing all points representing the complex numbers Z such that

3

$$|Z - 1 - i| \leq 1 \text{ and } -\frac{\pi}{4} \leq \operatorname{Arg}(Z - i) \leq \frac{\pi}{4}$$

ii) Let ϕ be the complex number of minimum modulus satisfying the inequalities of i).

1

Express ϕ in the form $x + yi$

d) Express $\phi = \frac{-1 + i}{\sqrt{3} + i}$ in modulus / argument form.

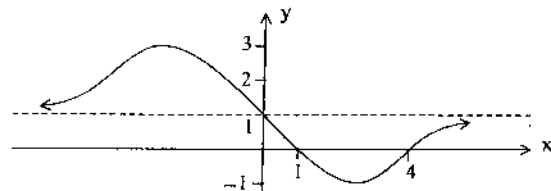
4

Hence, evaluate $\cos \frac{7\pi}{12}$ in surd form.

QUESTION 3 (Start a new page)

- a) Consider the equation $x^3 + 7x - 6i = 0$.
- Given that this equation has no purely real root, show that none of the roots is a conjugate of any of the others.
 - If $2i$ is one of the roots and the other two roots are purely imaginary, find the other two roots.

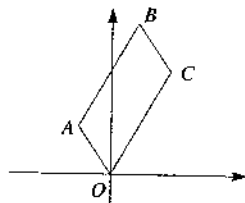
b)



The above diagram shows the graph of $y = f(x)$. Sketch on separate diagrams the following curves, indicating clearly any turning points and asymptotes.

- $y = \frac{1}{f(x)}$
- $y = f(|x|)$
- $y = \ln f(x)$
- $y = \sin^{-1}(f(x))$

c)



In the diagram above, $OABC$ is a parallelogram with $OA = \frac{1}{2}OC$.

The point A represents the complex number $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

If $\angle AOC = 60^\circ$, what complex number does C represent?

MARKS

1

2

2

2

2

3

3

QUESTION 4 (Start a new page)

MARKS

- a) Factorise $P(x) = x^4 - 5x^3 + 4x^2 + 2x - 8$ over
- $\mathbb{R}$ (all real numbers)
 - $\mathbb{C}$ (all complex numbers)

3

- b) Write down all polynomials that have degree 4, 3 as a single zero and -1 as a zero of multiplicity 3.

1

- c) If α, β, ρ are the roots of $x^3 - 2x^2 + x + 3 = 0$ evaluate:

(i) $\alpha^2 + \beta^2 + \rho^2$ (ii) $\alpha^3 + \beta^3 + \rho^3$

3

- d) If α, β, ρ are the roots of $x^3 + 2x^2 - 2x + 3 = 0$ form the equation whose roots are:

(i) $2\alpha, 2\beta, 2\rho$ (ii) $\alpha^2, \beta^2, \rho^2$

4

- e) The roots of the polynomial $P(x) = x^3 + ax^2 + bx + c = 0$ are in arithmetic progression. Show that the relationship between the coefficients of $P(x)$ is $2a^3 = 9ab - 27c$

3

- f) Prove that if α is a root of multiplicity r of $P(x)$ then it is a root of multiplicity $(r - 1)$ of $P'(x)$.

1

QUESTION 5 (Start a new page)

MARKS

- a) i) Show that the equation of the chord of contact of the tangents from a point (x_0, y_0) to the rectangular hyperbola $xy = c^2$ is $xy_0 + x_0y = 2c^2$.
- ii) Hence find the chord of contact of the tangents from the point $(2, 1)$ to the hyperbola $xy = 4$ and determine the points of contact.

5

- b) i) Show that the condition for the line $y = mx + c$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2 m^2 + b^2$.
- ii) Hence show that the pair of tangents from the point $(3, 4)$ to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ are at right angles to one another.

5

- c) i) Show that the equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$ is $a \sin \theta x + by = (a^2 + b^2) \tan \theta$.
- ii) The normal at the point $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the x-axis at G.

5

PN is the perpendicular from P to the x-axis.

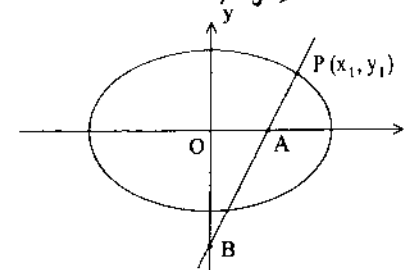
Prove that $OG = e^2 \cdot ON$, where O is the origin.

6

QUESTION 6 (Start a new page)

MARKS

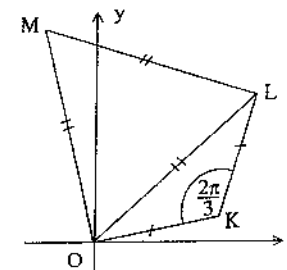
a)



The point $P(x_1, y_1)$, where $x_1 > 0$ and $y_1 > 0$, lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The normal at P intersects the x axis at A and the y axis at B.

- i) Show that the equation of the normal is $\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$ 3
- ii) Explain why the point A cannot be the focus of the ellipse. 2
- iii) Find the ratio in which A divides the interval BP internally. 2
- iv) Find the midpoint M of AB in terms of x_1 and y_1 . 1
- v) Given that H divides the interval OM in the ratio 4:1, show that the locus of H is an ellipse. 3

b)



The points K and M in a complex plane represent the complex numbers α and β respectively. The triangle OKL is isosceles and $\angle OKL = \frac{2\pi}{3}$. The triangle OLM is equilateral.

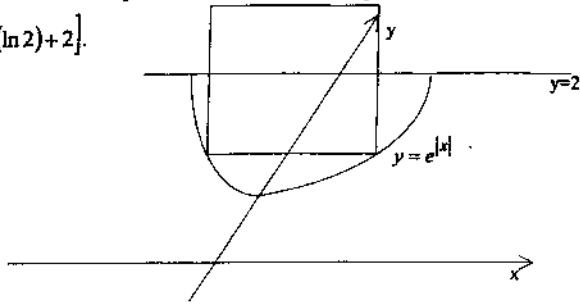
Show that $3\alpha^2 + \beta^2 = 0$

4

QUESTION 7 (Start a new page)

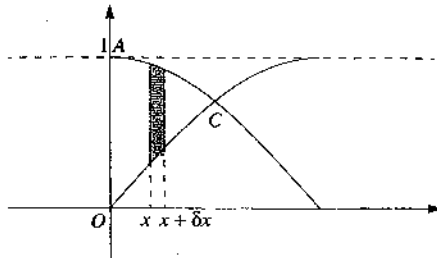
MARKS

- a) The base of a solid is formed by the segment cut off by the line $y = 2$ of the curve $y = e^{|x|}$. Cross sections of the solid parallel to the x axis are squares. Show that the volume is given by $4[2(\ln 2)^2 - 4(\ln 2) + 2]$.



8

- b) The diagram below shows part of the graphs of $y = \cos x$ and $y = \sin x$. The graph of $y = \cos x$ meets the y axis at A, and the C is the first point of intersection of the two graphs to the right of the y axis.



The region OAC is to be rotated about the line $y = 1$.

- (i) Write down the coordinates of the point C. 1
- (ii) The shaded strip of width δx shown in the diagram is rotated about the line $y = 1$. Show that the volume δV of the resulting slice is given by $\delta V = \pi(2\cos x - 2\sin x + \sin^2 x - \cos^2 x)\delta x$. 3
- (iii) Hence evaluate the total volume when the region OAC is rotated about the line $y = 1$. 3

QUESTION 8 (Start a new page)

MARKS

- a) Let $I_n = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^n x \, dx$, where n is a positive integer.

- i) Using integration, show that

4

$$(n-1) I_n = 2^{n-2} \sqrt{3} + (n-2) I_{n-2}$$

- ii) Evaluate $J = \int_0^{\frac{\pi}{3}} \sec^4 x \, dx$

3

- b) Consider the polynomial $x^5 - i = 0$

- i) Show that $1 - ix - x^2 + ix^3 + x^4 = 0$ for $x \neq i$

2

- ii) Show that

$$(x-i) \left(x^2 - 2i \sin \frac{\pi}{10} x - 1 \right) \left(x^2 + 2i \sin \frac{3\pi}{10} x - 1 \right) = 0$$

4

- iii) Show that $\sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4}$

2

IGS - Year 12 Trial (4U) Extension 2 Solutions.

Question 1

$$1) \int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} dx$$

$$\text{let } u = \tan x \\ du = \sec^2 x dx$$

$$= \int \frac{du}{\sqrt{1-u^2}}$$

$$= \sin^{-1} u + C$$

$$= \sin^{-1}(\tan x) + C$$

$$2) \frac{1}{(x^2+4)(2-x)} = \frac{ax+b}{x^2+4} + \frac{c}{2-x}$$

$$16 = (ax+b)(2-x) + c(x^2+4)$$

$$\text{then } x=2$$

$$16 = 16c \quad \therefore c = 1$$

$$16 = 16$$

$$\text{equating coefficients of } x^2$$

$$-a+c=0$$

$$c=a$$

$$2+2$$

$$\text{equating coefficients of } x$$

$$2a-b=0$$

$$b=2a$$

$$b=4$$

$$\int \frac{16}{(x^2+4)(2-x)} dx = \int \left(\frac{4}{x^2+4} + \frac{2}{2-x} \right) dx$$

$$I = \int \left(\frac{2x}{x^2+4} + \frac{4}{x^2+4} + \frac{2}{2-x} \right) dx \\ = \ln|x^2+4| + \frac{4}{2} \tan^{-1} \frac{x}{2} - 2 \ln|2-x| + C \\ = 2 \tan^{-1} \frac{x}{2} + \ln \left(\frac{x^2+4}{(2-x)^2} \right) + C$$

$$c) \int \frac{\ln x}{x^2} dx$$

$$\text{let } u = \ln x$$

$$v = \frac{1}{x^2}$$

$$u' = \frac{1}{x}$$

$$v' = -\frac{1}{x^3}$$

$$\therefore I = uv - \int v u' \\ = \ln x \left(-\frac{1}{x} \right) + \int \frac{1}{x^3} dx \\ = -\frac{\ln x}{x} - \frac{1}{2x^2} + C$$

$$d) \int_0^{\frac{\pi}{2}} \frac{d\theta}{4 \cos \theta - 3 \sin \theta + 6}$$

$$\text{let } t = \tan \frac{\theta}{2} \quad \frac{d\theta}{2} = \frac{sec^2 \frac{\theta}{2}}{1+t^2} dt$$

$$\cos \theta = \frac{1-t^2}{1+t^2} \quad \frac{2dt}{1+t^2} = d\theta$$

$$\sin \theta = \frac{2t}{1+t^2}$$

$$I = \int_0^1 \frac{2dt}{4 \left(\frac{1-t^2}{1+t^2} \right) - 3 \left(\frac{2t}{1+t^2} \right) + 6} \\ = \int_0^1 \frac{2dt}{8t - 2 + 2t^2 + 6 + 6t^2} \\ = \int_0^1 \frac{2dt}{8t^2 - 2 + 6t^2}$$

Question 1 (continued)

$$I = \int_0^1 \frac{dt}{4t^2 + 4t + 2}$$

$$= \int_0^1 \frac{dt}{(2t+1)^2 + 1}$$

$$= \left[\frac{1}{2} \tan^{-1} (2t+1) \right]_0^1$$

$$= \frac{1}{2} \left(\tan^{-1} 3 - \tan^{-1} 1 \right)$$

$$\therefore \tan^{-1} 3 - \tan^{-1} 1 = \frac{\pi}{4}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\frac{3-1}{1+3 \cdot 1} = \frac{2}{4} = \frac{1}{2}$$

$$\tan(A-B) = \frac{1}{2}$$

$$A-B = \tan^{-1} \left(\frac{1}{2} \right)$$

$$I = \frac{1}{2} \tan^{-1} \frac{1}{2}$$

$$= \frac{1}{2} \tan^{-1} \frac{1}{2}$$

$$a) \text{ let } z = x + iy$$

$$\bar{z} = x - iy$$

$$\frac{1}{z-1} = \frac{1}{x-1+iy} \cdot \frac{x-1-iy}{x-1-iy}$$

$$= \frac{x-1-iy}{(x-1)^2 + y^2}$$

$$\int \left(\frac{1}{z-1} \right) dz = 1$$

$$\therefore \frac{y+1}{x^2+(y-1)^2} = 1$$

$$y+1 = x^2 + (y-1)^2$$

$$x^2 + y^2 - 2y = 0$$

$$x^2 + (y-1)^2 = 1$$

The locus is a circle centre $(0, 1)$, radius 1

$$b) \text{ let } (5+12i)^n = x + iy$$

$$5^n + 12^n = x^2 + y^2$$

equating coefficients of imaginary parts

$$x^2 - y^2 = 1 \text{ and } -12 = 2xy$$

$$\frac{x}{y} = \frac{1}{2} \text{ or } -2$$

and since $x^2 + y^2 = 1$

$$\frac{x}{y} = \frac{1}{2} \Rightarrow x = \frac{y}{2}$$

$$\frac{y^2}{4} + y^2 = 1$$

$$\frac{5y^2}{4} = 1 \Rightarrow y^2 = \frac{4}{5}$$

$$y = \pm \frac{2}{\sqrt{5}}$$

$$x = \pm \frac{1}{\sqrt{5}}$$

therefore $x = \pm \frac{1}{\sqrt{5}}$

and $y = \pm \frac{2}{\sqrt{5}}$

$$\therefore \sqrt{5} = 3-2i \text{ or } -3+2i$$

Question 2 (continued)

(b) (i) $z = \frac{1 + \sqrt{5-12i}}{2+2i}$

$$Z = \frac{1+3-2i}{2+2i} \quad \text{or} \quad Z = \frac{1-3+2i}{2+2i}$$

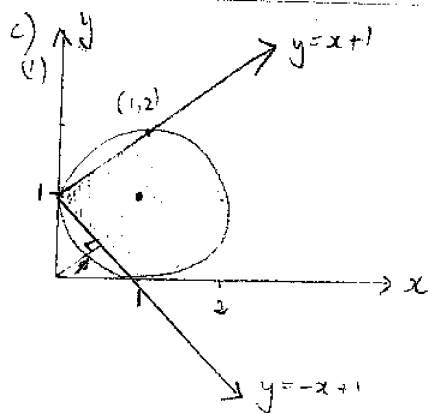
$$Z = \frac{4-2i}{2+2i} \quad \text{or} \quad Z = \frac{-2+2i}{2+2i}$$

$$Z = \frac{2-i}{1+i} \times \frac{1-i}{1-i} \quad \text{or} \quad Z = \frac{-1+i}{1+i} \times \frac{1-i}{1-i}$$

$$Z = \frac{1-3i}{2} \quad \text{or} \quad Z = i$$

choose $Z = i$ (as it is purely imaginary)

$$Z^{100} = i^{100} = (i^4)^{25} = 1$$

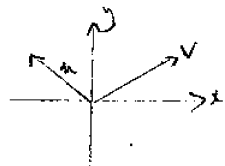


(ii) ϕ is complex number of minimum modulus.
 $\therefore$ shortest distance to line $y = -x + 1$ is the complex number $\phi = \frac{1}{2} + \frac{1}{2}i$

d) $\phi = \frac{-1+i}{\sqrt{3}+i} = \frac{u}{v}$

$$\text{let } u = -1+i = \sqrt{2} \cos \frac{3\pi}{4}$$

$$v = \sqrt{3}+i = 2 \cos \frac{\pi}{6}$$



$$\therefore \phi = \frac{\sqrt{2}}{2} \cos \left(\frac{3\pi}{4} - \frac{\pi}{6} \right)$$

$$\phi = \frac{1}{\sqrt{2}} \cos \frac{7\pi}{12} \quad (\text{in mod-arg form})$$

$$\text{If } \phi = \frac{-1+i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i}$$

$$\frac{1-\sqrt{3}+i(1+\sqrt{3})}{4}$$

equating real parts of ϕ

$$\frac{1}{\sqrt{2}} \cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{4}$$

$$\cos \frac{7\pi}{12} = \frac{\sqrt{2}-\sqrt{6}}{4}$$

Question 3

a) (i) For the equation $x^3 + 7x - 6i = 0$

Let α, β, γ be the roots

$$\text{Sum of roots } \alpha + \beta + \gamma = 0$$

$$\text{If we let } \alpha = x + iy \quad \text{are conjugates}$$

$$\beta = x - iy$$

$$\gamma = a + ib$$

$$\therefore x + \beta + \gamma = 2x + a + ib = 0$$

$$-2 \quad b = 0$$

If none of the roots are purely real, then we can't assume the roots are conjugates.

(ii) Let $\alpha = 2i$
 $\beta = 3i$
 $\gamma = -i$

then sum of roots

$$i(2+3-1) = 0$$

$$\therefore b + c = -2 \quad \text{--- (1)}$$

product of roots

$$-i(2 \cdot 3) = 6i$$

$$bc = -3 \quad \text{--- (2)}$$

sub (2) into (1)

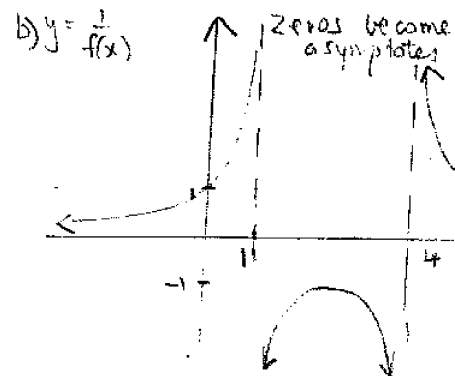
$$-3 + c = -2$$

$$c + c = 3 \Rightarrow c = 0$$

$$b = -1 \text{ or } b = 3$$

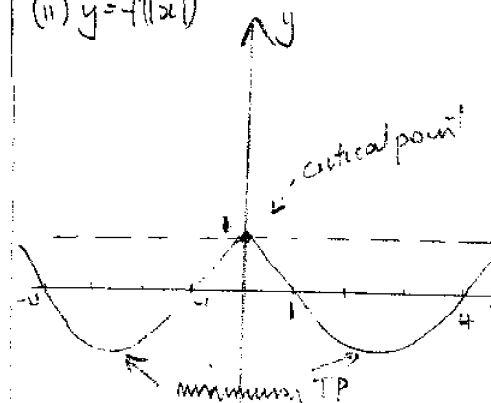
$$c = 1 \text{ or } c = 2$$

$\therefore$ roots are $x = 2i, x = 3i, x = -i$



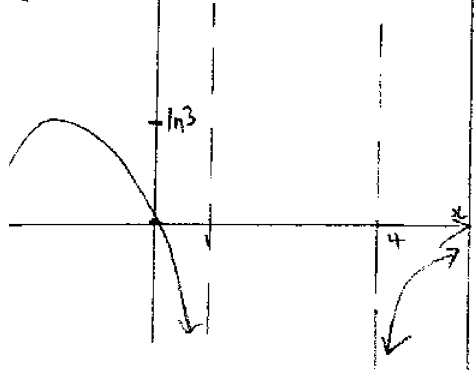
max becomes min
 increasing becomes decreasing

(ii) $y = f(f(x))$

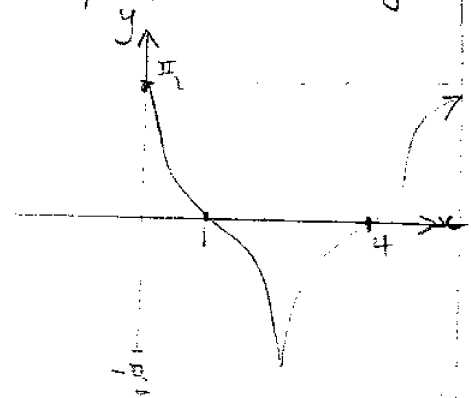


Positive x values are reflected about the y-axis

$y = \ln f(x)$ is defined only for $f(x) > 0$



1) $y = \sin^{-1}(f(x))$ is only defined for $-1 \leq y \leq 1$



$$c) |OA| = \frac{1}{2}|OC|$$

$$|OC| = 2\left(\frac{1}{4} + \frac{3}{4}\right) = 2$$

$$\arg(OA) = \tan^{-1} \frac{\sqrt{3}}{-1} = 120^\circ$$

$$\therefore \angle COA = 60^\circ$$

$$\therefore C = 2(\cos 60^\circ + i \sin 60^\circ)$$

$$= 2 \times \frac{1}{2} + i 2 \times \frac{\sqrt{3}}{2}$$

$$\underline{C = 1 + \sqrt{3}i}$$

Question 4

a) $P(x) = x^4 - 5x^3 + 4x^2 + 2x - 8$

$$P(-1) = 1 + 5 + 4 - 2 - 8 = 0$$

$\therefore (x+1)$ is a factor

$$\begin{array}{r} x^3 - 6x^2 + 10x - 8 \\ x+1 \overline{) x^4 - 5x^3 + 4x^2 + 2x - 8} \\ \underline{x^4 + x^3} \\ -6x^3 + 4x^2 \\ \underline{-6x^3 + 6x^2} \\ 10x^2 + 10x - 8 \\ \underline{10x^2 + 10x} \\ -8x - 8 \\ \underline{-8x - 8} \\ 0 \end{array}$$

$$Q(x) = x^3 - 6x^2 + 10x - 8$$

$$Q(4) = 64 - 96 + 40 - 8 = 0$$

$\therefore (x-4)$ is a factor

$$\begin{array}{r} x^2 - 2x + 2 \\ x-4 \overline{) x^3 - 6x^2 + 10x - 8} \\ \underline{x^3 - 4x^2} \\ -2x^2 + 10x \\ \underline{-2x^2 + 8x} \\ 2x - 8 \\ \underline{2x - 8} \\ 0 \end{array}$$

$$S(x) = x^2 - 2x + 2$$

$\therefore$ over $\mathbb{R}$

$$P(x) = (x+1)(x-4)(x^2 - 2x + 2)$$

over $\mathbb{R} \quad x = 2 \pm \sqrt{-1}$

$$= 2 \pm i$$

$P(x) = (x-4)(x+1)(x-1-i)(x-1+i)$
over $\mathbb{C}$

b) $P(x) = k(x+1)^3(x-3)$

c) $\alpha + \beta + \gamma = 2$

$$\alpha\beta + \beta\gamma + \alpha\gamma = 1$$

$$\alpha\beta\gamma = 3$$

$$\therefore \alpha^4 + \beta^4 + \gamma^4 = (\alpha^2 + \beta^2 + \gamma^2)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= 2^2 - 2(1)$$

$$= 2$$

(ii) α, β, γ are the roots of $x^3 - 2x^2 + x + 3 = 0$

then $\alpha^3 = 2\alpha^2 + \alpha + 3$

$$\beta^3 = 2\beta^2 + \beta + 3$$

$$\gamma^3 = 2\gamma^2 + \gamma + 3$$

$$\therefore \alpha^4 + \beta^4 + \gamma^4 = 2(\alpha^3 + \beta^3 + \gamma^3) + (\alpha^2 + \beta^2 + \gamma^2)$$

$$= 2(2) + 2 + 9$$

$$= 15$$

d) $x^3 + 3x^2 - 2x + 3 = 0$

$$\left(\frac{x}{2}\right)^3 + 3\left(\frac{x}{2}\right)^2 + 2\left(\frac{x}{2}\right) + 3 = 0$$

$$\frac{x^3}{8} + \frac{3x^2}{4} + \frac{x}{2} + 3 = 0$$

$$y^3 + 4y^2 + 8y + 24 = 0$$

$$x^3 + 2x^2 + 2x + 3 = 0$$

$$(\sqrt{y})^3 + 2(\sqrt{y})^2 + 2\sqrt{y} + 3 = 0$$

$$y\sqrt{y} + 2\sqrt{y} = -2y - 3$$

$$\sqrt{y}(y+2) = -2y-3$$

$$y(y^2+4y+4) = 4y^2+12y+9$$

$$y^3+4y^2+4y = 4y^2+12y+9$$

$$y^3 - 8y^2 + 8y - 9 = 0$$

$$f) P(x) = (x-\alpha)^r Q(x)$$

$$P'(x) = r(x-\alpha)^{r-1}Q(x) + Q'(x)(x-\alpha)$$

$$= (x-\alpha)^{r-1} (rQ(x) + Q'(x)(x-\alpha))$$

$$P'(x) = (x-\alpha)^{r-1} S(x)$$

$\therefore$ if we want to multiply

$$(x-\alpha)^r \cdot P'(x)$$

Question 5

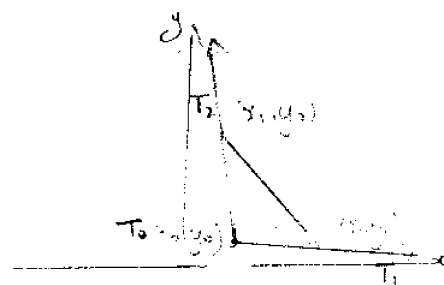
$$2) i) y = \frac{c}{x}$$

$$\frac{dy}{dx} = -\frac{c}{x^2} = -\frac{y}{x}$$

at a point (x_1, y_1)

$$\frac{dy}{dx} = -\frac{y_1}{x_1}$$

at a point $(x_2, y_2) \frac{dy}{dx} = -\frac{y_2}{x_2}$



$$\text{Eqn of } T_1: (y-y_1) = -\frac{y_1}{x_1}(x-x_1)$$

$$\text{Eqn of } T_2: (y-y_2) = -\frac{y_2}{x_2}(x-x_2)$$

$$T_1: x_1y + y_1x = 2\frac{y_1}{x_1} = 2c^2$$

$$T_2: x_2y + y_2x = 2c^2$$

Question 5 (continued)

i) The tangents T_1 and T_2

intersect at $T_0(x_0, y_0)$

$$\therefore x_1y_0 + y_1x_0 = 2c^2$$

$$\therefore x_2y_0 + y_2x_0 = 2c^2$$

and hence (x_1, y_1) and (x_2, y_2)

$$\text{satisfy } x_0y + y_0x = 2c^2$$

ii) The chord of contact to the hyperbola $xy = 4$ at $(2, 1)$ has equation

$$2y + x = 8 \quad (\text{from above})$$

This chord of contact intersects $y = \frac{4}{x}$ when

$$2(\frac{4}{x}) + x = 8$$

$$8 + x^2 = 8x$$

$$x^2 - 8x + 8 = 0$$

$$x = \frac{8 \pm \sqrt{64 - 32}}{2}$$

$$= \frac{8 \pm \sqrt{32}}{2}$$

$$x = 4 \pm 2\sqrt{2}$$

$$\text{when } x = 4 + 2\sqrt{2} \quad y = \frac{4}{4+2\sqrt{2}}$$

$$x = 4 - 2\sqrt{2} \quad y = \frac{4}{4-2\sqrt{2}}$$

$\therefore$ pts of contact are:

$$(4+2\sqrt{2}, \frac{2}{2+\sqrt{2}}) (4-2\sqrt{2}, \frac{2}{2-\sqrt{2}})$$

$$\text{rationalising denominator} \\ (\frac{2}{2+\sqrt{2}} \cdot \frac{2-\sqrt{2}}{2-\sqrt{2}} = \frac{2-2\sqrt{2}}{2} = 2-\sqrt{2})$$

$$(4+2\sqrt{2}, 2-\sqrt{2}) (4-2\sqrt{2}, 2+\sqrt{2})$$

$$b) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

ii) If $y = mx + c$ is a tangent to the ellipse then on substitution there should only be one solution i.e. $\Delta = 0$

$$\frac{x^2}{a^2} + \frac{(mx+c)^2}{b^2} = 1$$

$$b^2x^2 + a^2(m^2x^2 + 2mcx + c^2) = a^2 \\ x^2(b^2 + a^2m^2) + x(2a^2mc) + a^2(c^2 - b^2) = 0$$

if $\Delta = 0$, then

$$(2a^2mc)^2 - 4(b^2 + a^2m^2)a^2(c^2 - b^2) = 0 \\ 4a^4m^2c^2 - 4a^2b^2c^2 - 4a^4m^2c^2 + 4a^2b^2 + 4a^4m^2c^2 = 0$$

$$4a^2b^2c^2 = 4a^2b^2 + 4a^4m^2$$

$$(\div 4a^2b^2) \\ c^2 = b^2 + a^2m^2 \\ c^2 = a^2m^2 + b^2 \quad \text{geo}$$

$$b^2 = a^2(e^2 - 1) \quad \frac{b^2}{a^2} = e^2 - 1$$

Question 5 (continued)

$$(c) \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\text{has gradient } \frac{dy}{dx} = \frac{x b^2}{y a^2}$$

$$\text{at } (a \sec \theta, b \tan \theta)$$

$$\frac{dy}{dx} = \frac{a \sec \theta b^2}{b \tan \theta a^2}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

$$= \frac{b \frac{1}{\cos \theta}}{a \frac{\sin \theta}{\cos \theta}}$$

$$\frac{dy}{dx} = \frac{b}{a \sin \theta}$$

$$\therefore \text{gradient of normal is } -\frac{a \sin \theta}{b}$$

Eqn of normal is

$$y - b \tan \theta = -\frac{a \sin \theta}{b} (x - a \sec \theta)$$

$$by - b^2 \tan \theta = -a \sin \theta x + a^2 \tan \theta$$

$$a \sin \theta x + by = (a^2 + b^2) \tan \theta$$

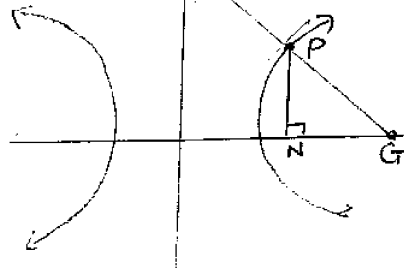
(i) at G, y = 0

$$\therefore x = \frac{(a^2 + b^2) \tan \theta}{a \sin \theta}$$

$$x = \frac{a^2 + b^2}{a \cos \theta} \quad G = \left(\frac{a^2 + b^2}{a \cos \theta}, 0 \right)$$

at N, y = 0

$$N = (a \sec \theta, 0)$$



$$OG = \frac{a^2 + b^2}{a \cos \theta}$$

$$ON = a \sec \theta = \frac{a}{\cos \theta}$$

$$OG = \frac{a^2 + b^2}{a}, \quad ON = \frac{a}{\cos \theta}$$

$$OG = \frac{a^2 + b^2}{a^2} \cdot ON$$

$$\text{where } b^2 = a^2(e^2 - 1)$$

$$\therefore \frac{a^2 + b^2}{a^2} = \frac{a^2 + a^2(e^2 - 1)}{a^2}$$

$$= \frac{a^2(1 + e^2 - 1)}{a^2} = e^2$$

$$\therefore OG = e^2 \cdot ON$$

Question 6

$$a) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\text{at } (x_1, y_1)$$

$$\frac{dy}{dx} = -\frac{b^2 x_1}{a^2 y_1}$$

$$\text{grad of normal is } \frac{a^2 y_1}{b^2 x_1}$$

Equation of normal is

$$(y - y_1) = \frac{a^2 y_1}{b^2 x_1} (x - x_1)$$

$$b^2 x_1 y - b^2 y_1 x_1 = a^2 y_1 x - a^2 y_1 x_1$$

$$(a^2 - b^2) x_1 y_1 = a^2 x y_1 - b^2 y x_1$$

$$a^2 - b^2 = a^2 \frac{x}{x_1} - b^2 \frac{y}{y_1}$$

$$\therefore \frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2$$

(In the equation of the normal)

At A, on the x-axis, y = 0.

$$\therefore \frac{a^2 x}{x_1} = a^2 - b^2 = a^2 e^2$$

$$x = \frac{a^2 - b^2}{a^2} x_1 = e^2 x_1$$

If A is the focus, then x = ae

$$\therefore e^2 x_1 = ae \quad \therefore x_1 = \frac{a}{e}$$

$\therefore \frac{a}{e}$ is the equation of directrix, which does not lie on the ellipse.
 $\therefore A$ is not the focus

(ii) at B, x = 0.

$$\therefore y = \frac{a^2 - b^2}{b^2} y_1 = \frac{a^2 e^2}{b^2} y_1$$

$$\therefore B(0, -\frac{a^2 e^2 y_1}{b^2})$$

$$\text{and } A(x_1 e^2, 0)$$

now A divides BP in the ratio m:n where

$$x_1 e^2 = \frac{m x_1 + n \cdot 0}{m+n}$$

$$\text{not required } 0 = \frac{m y_1 + n (-\frac{a^2 e^2 y_1}{b^2})}{m+n}$$

$$\text{from } \odot \quad e^2(m+n) = m$$

$$e^2 n = m(1 - e^2)$$

$$\frac{e^2}{1 - e^2} = \frac{m}{n}$$

$$\therefore m:n = e^2 : 1 - e^2$$

$$A(x_1 e^2, 0) \quad B(0, -\frac{a^2 e^2 y_1}{b^2})$$

$$\therefore M(\frac{e^2 x_1}{2}, -\frac{a^2 e^2 y_1}{2 b^2})$$

Question 6 (continued)

(c) H divides OM in the ratio 4:1

$$= \left(\frac{4 \left(\frac{x_1 e^2}{a} \right) + 1(0)}{5}, \frac{4 \left(\frac{-a^2 e^2 y_1}{2b^2} \right) + 1(0)}{5} \right)$$

$$= \left(\frac{2e^2 x_1}{5}, \frac{-2a^2 e^2 y_1}{5b^2} \right)$$

$$= (X, Y)$$

to the above we find a relationship between X and Y

$$\frac{5X}{2e^2} = x_1 \quad \frac{5Yb^2}{-2a^2 e^2} = y_1$$

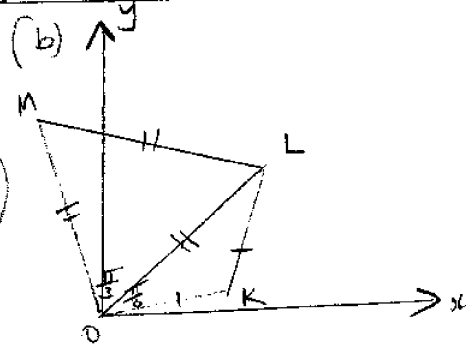
as (x_1, y_1) lies on the ellipse then $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$

$$\frac{25X^2}{4a^2 e^4} + \frac{25Y^2 b^2}{4a^4 e^4} = 1$$

$$+ (b^2 = a^2(1-e^2)) \text{ and } ab^2 = a^2 b^2$$

$$\frac{X^2}{\frac{4}{25} \left(\frac{a^2 b^2}{a^2} \right)^2} + \frac{Y^2}{\frac{4}{25} \left(\frac{a^2 b^2}{b^2} \right)^2}$$

this is the locus of an ellipse



$$\text{given } \angle OKL = \frac{2\pi}{3}$$

$$\therefore \angle OKL = \angle LOK = \frac{\pi - \frac{2\pi}{3}}{2} = \frac{\pi}{6}$$

$$\angle LOM = \frac{\pi}{3} \quad (\triangle OML \text{ is equilateral})$$

$$\text{now } \angle KOM \text{ is } \frac{\pi}{2}$$

$\therefore$ OK to OM is a rotation through $\frac{\pi}{2}$ and an enlargement by k .

To find k ,

$$|OK| = |OL| = |LK|$$

$$\text{and } |OL| = \sqrt{3}|OK|$$

$$\therefore k = \sqrt{3}$$

let α is rotated through $\frac{\pi}{2}$

means a multiple of $\frac{\pi}{2}$

$$\therefore \beta = \alpha \pm \frac{\pi}{2}$$

convert both sides

$$\beta^2 = -3\alpha^2$$

$$3\alpha^2 + \beta^2 = 0 \quad \text{qed}$$

Question 7

$$a) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow y^2 = \frac{a^2 b^2 - b^2 x^2}{a^2} = b^2 - \frac{b^2}{a^2} x^2$$

consider slices $\perp$ to x-axis

$$\begin{aligned} A &= \pi((c+y)^2 - (c-y)^2) \\ &= \pi(c^2 + 2cy + y^2 - c^2 + 2cy - y^2) \\ &= 4\pi cy \end{aligned}$$

$$\therefore V = 4\pi cy \, dx$$

$$V = 2 \int_0^a \partial V$$

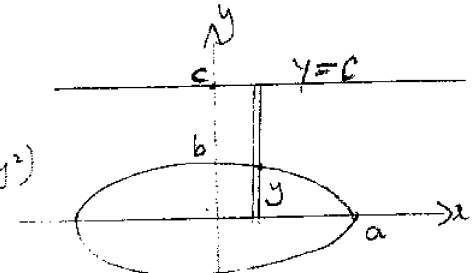
$$= 8\pi \int_0^a cy \, dx$$

$$= 8\pi c \int_0^a \sqrt{\frac{a^2 b^2 - b^2 x^2}{a^2}} \, dx$$

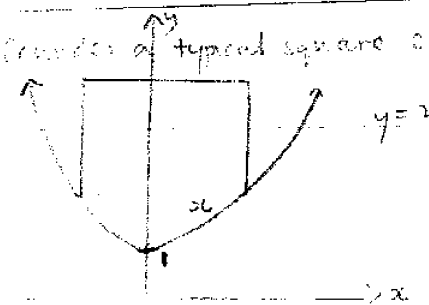
$$= \frac{8\pi c b}{a} \int_0^a \underbrace{\sqrt{a^2 - x^2}}_{\frac{1}{4} \text{ circle}} \, dx$$

$$= \frac{8\pi c b}{a} \cdot \frac{1}{4} \pi a^2$$

$$= 2\pi abc \text{ units}^3$$



b) Consider a typical square cross-section Area = $(2|x|)^2$
 $A = 4x^2$



$$\partial V = 4x^2 \, dy$$

$$V = \int_1^2 4x^2 \, dy$$

$$\begin{aligned} y &= e^{|x|} \\ \log_e y &= |x| \\ (|x|)^2 &= x^2 \end{aligned}$$

Question 7

$$b) V = 4 \int_1^2 (\ln y)^2 \cdot dy$$

$$\text{let } u' = 1 \quad v = (\ln y)^2 \\ u = y \quad v' = 2(\ln y) \cdot \frac{1}{y}$$

$$V = 4 \left[y (\ln y)^2 \right]_1^2 - \int_1^2 \frac{2}{y} \ln y \cdot y \cdot dy$$

$$= 8(\ln 2)^2 - 8 \int_1^2 \ln y \cdot dy$$

$$\text{let } u = \ln y \quad v' = 1 \\ u' = \frac{1}{y} \quad v = y$$

$$V = 8(\ln 2)^2 - 8 \left[y \ln y \right]_1^2 - \int_1^2 y \cdot \frac{1}{y} \cdot dy$$

$$= 8(\ln 2)^2 - 16(\ln 2) + 8[y]_1^2$$

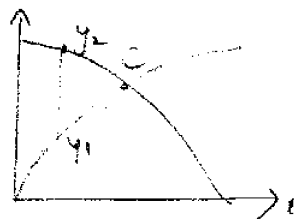
$$= 4(2(\ln 2)^2 - 2(\ln 2) + 2) \quad \text{qed}$$

$$c) y = \cos x \text{ and } y = \sin x$$

$$\cos x = \sin x$$

$$\tan x = 1$$

$$x = \frac{\pi}{4} \quad \text{Hence } C = \left(\frac{\pi}{4}, \frac{1}{\sqrt{2}} \right)$$



$$(ii) \partial V = \pi [(1-y_1)^2 - (1-y_2)^2] \partial x$$

$$= \pi [(1-\sin x)^2 - (1-\cos x)^2] \partial x$$

$$= \pi (1 - 2\sin x + \sin^2 x - 1 + 2\cos x - \cos^2 x) \partial x$$

$$\partial V = \pi (2\cos x - 2\sin x + \sin^2 x - \cos^2 x) \cdot \partial x$$

qed

$$(iii) V = \lim_{\partial x \rightarrow 0} \sum_{x=0}^{\frac{\pi}{4}} \pi (2\cos x - 2\sin x + \sin^2 x - \cos^2 x) \cdot \partial x$$

$$= \pi \int_0^{\frac{\pi}{4}} (2\cos x - 2\sin x + \sin^2 x - \cos^2 x) dx$$

$$= \pi \int_0^{\frac{\pi}{4}} (2\cos x - 2\sin x - \cos 2x) dx$$

$$= \pi \left(2\sin x + 2\cos x - \frac{\sin 2x}{2} \right) \Big|_0^{\frac{\pi}{4}}$$

$$= \pi \left(\left(2\frac{1}{\sqrt{2}} + 2\frac{1}{\sqrt{2}} - \frac{1}{2} \right) - (0 + 2 - 0) \right)$$

$$= \pi (2\sqrt{2} - \frac{5}{2})$$

$$= \frac{\pi}{2} (4\sqrt{2} - 5) \text{ cub units.}$$

Question 8:

$$\begin{aligned} a) I_n &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^n x \cdot dx \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^2 x \operatorname{cosec}^{n-2} x \cdot dx \end{aligned}$$

$$\text{let } u' = \operatorname{cosec}^2 x$$

$$u = -\cot x$$

$$v = \operatorname{cosec}^{n-2} x$$

$$v' = (n-2) \operatorname{cosec}^{n-3} x \cdot (-\operatorname{cosec} x \cot x)$$

$$v' = -(n-2) \operatorname{cosec}^{n-2} x \cot x$$

$$I_n = -\cot x \operatorname{cosec}^{n-2} x \Big|_{\frac{\pi}{6}}^{\frac{\pi}{2}} - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (n-2) \operatorname{cosec}^{n-2} x \cot^2 x \cdot dx$$

$$\text{but } \cot^2 x + 1 = \operatorname{cosec}^2 x$$

$$\text{and } \cot \frac{\pi}{2} = 0$$

$$\cot \frac{\pi}{6} = \sqrt{3}$$

$$\operatorname{cosec} \frac{\pi}{6} = 2$$

$$I_n = \sqrt{3} (2)^{n-2} - (n-2) \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^{n-2} x (\operatorname{cosec}^2 x - 1) \cdot dx$$

$$I_n = \sqrt{3} (2)^{n-2} - (n-2) \underbrace{\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^n x \cdot dx}_{I_n} + (n-2) \underbrace{\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^{n-2} x \cdot dx}_{I_{n-2}}$$

$$n(1+n-2) = \sqrt{3} (2)^{n-1} + (n-2) I_{n-2}$$

$$(n-1) I_n = \sqrt{3} (2)^{n-1} + (n-2) I_{n-2} \quad \text{qed}$$

Question 8 (continued)

$$i) J = \int_0^{\frac{\pi}{3}} \sec^4 x \cdot dx$$

$$\text{using } \sec x = \operatorname{cosec}(\frac{\pi}{2} - x)$$

$$\text{let } u = \frac{\pi}{2} - x \\ du = -dx$$

$$J = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \operatorname{cosec}^4 u \cdot du = I_4$$

$$3I_4 = 2^2 \sqrt{3} + 2I_2$$

$$I_2 = \sqrt{3}$$

$$\therefore 3I_4 = 4\sqrt{3} + 2\sqrt{3}$$

$$I_4 = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$$

$$\therefore J = 2\sqrt{3}$$

$$b) x^5 - i = 0$$

$$i) (x-i)(x^4 + ix^3 + i^2 x^2 + i^3 x + i^4) = 0$$

$$(x-i)(x^4 + ix^3 - x^2 - ix + 1) = 0$$

$$\text{but since } x \neq i \text{ then}$$

$$x^4 + ix^3 - x^2 - ix + 1 = 0$$

$$ii) x^5 = i \quad \text{let } x = r \operatorname{cis} \theta$$

$$r^5 \operatorname{cis} 5\theta = i \quad (\text{by de Moivre's theorem})$$

$$r^5 \operatorname{cis} 5\theta = \operatorname{cis} \frac{\pi}{2} \quad \Rightarrow r = 1$$

$$\operatorname{cis} 5\theta = \operatorname{cis} \frac{\pi}{2}$$

$$5\theta = \frac{\pi}{2} + 2k\pi, \quad \theta = \frac{\pi}{10} + \frac{2k\pi}{5}$$

Question 3 (continued)

$$\theta = \frac{\pi}{10} + 2k\pi \quad k=0,1,2,3,4$$

when $k=0$, $\theta = \frac{\pi}{10} = \frac{\pi}{10}$

$k=1$, $\theta = \frac{\pi}{10} + \frac{2\pi}{5} = \frac{5\pi}{10} = \frac{\pi}{2}$

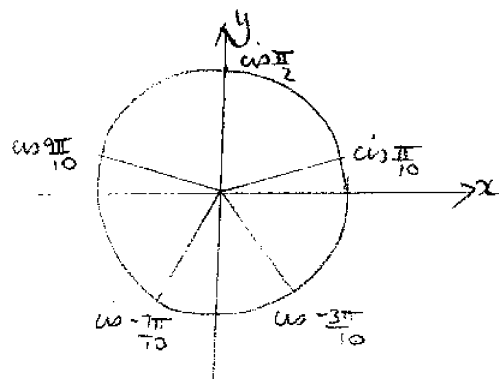
$k=2$, $\theta = \frac{\pi}{10} + \frac{4\pi}{5} = \frac{9\pi}{10}$

$k=3$, $\theta = \frac{\pi}{10} + \frac{6\pi}{5} = \frac{13\pi}{10} = \frac{3\pi}{2}$

$k=4$, $\theta = \frac{\pi}{10} + \frac{8\pi}{5} = \frac{17\pi}{10} = -\frac{3\pi}{10}$

$\therefore x^5 - 1 = 0$ can be expressed as

$$-a \cdot \frac{\pi}{10} (x - \cos \frac{\pi}{10}) (x - \cos \frac{9\pi}{10}) (x - \cos \frac{3\pi}{10}) (x - \cos \frac{5\pi}{10}) = 0$$



from the diagram let $\alpha = \cos \frac{\pi}{10}$

$$-\alpha = \cos \frac{9\pi}{10}$$

$$\text{let } \beta = \cos \frac{3\pi}{10}$$

$$-\beta = \cos \frac{7\pi}{10}$$

hence

$$(x-i)(x-\alpha)(x+\alpha)(x-\beta)(x+\beta) = 0$$

$$(x-i)(x^2+(\alpha-\alpha)x-\alpha\alpha)(x^2+(\beta-\beta)x-\beta\beta)$$

$$\text{but } \alpha\alpha = 1 \quad \beta\beta = 1$$

$$\bar{\alpha} - \alpha = \cos \frac{9\pi}{10} - \cos \frac{\pi}{10}$$

$$\bar{\alpha} - \alpha = \cos \frac{\pi}{10} - i \sin \frac{\pi}{10} - \cos \frac{\pi}{10} - i \sin \frac{\pi}{10}$$

$$\bar{\alpha} - \alpha = -2i \sin \frac{\pi}{10}$$

similarly

$$\bar{\beta} - \beta = -2i \sin \left(\frac{3\pi}{10} \right) = -2i \sin \frac{3\pi}{10}$$

Altogether,

$$(x-i)(x^2-2i \sin \frac{\pi}{10} x - 1)(x^2+2i \sin \frac{3\pi}{10} x - 1)$$

To show

$$(iii) \sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4}$$

we equate coefficients of x^2

Given,

$$(x^2-2i \sin \frac{\pi}{10} x - 1)(x^2+2i \sin \frac{3\pi}{10} x - 1)$$

$$= x^4 + ix^3 - x^2 - ix + 1$$

$$-1 + (2i \sin \frac{\pi}{10})(2i \sin \frac{3\pi}{10}) + 1 = -1$$

$$4i^2 \sin \frac{\pi}{10} \sin \frac{3\pi}{10} = 1$$

$$\sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4}$$